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global_data = {
    "attribute_1": "total_cost",
    "attribute_2": "material_continuity",
    "attribute_3": "thermal_performance",
    "attribute_4": "mass",
    "attribute_5": "corrosion_resistance",

    "direction_attribute_1": "negative",
    "direction_attribute_2": "positive",
    "direction_attribute_3": "positive",
    "direction_attribute_4": "negative",
    "direction_attribute_5": "positive",

    "attribute_1_min": 2000,
    "attribute_1_max": 50000,
    "attribute_2_min": 0,
    "attribute_2_max": 1,
    "attribute_3_min": 20,
    "attribute_3_max": 150,
    "attribute_4_min": 100,
    "attribute_4_max": 2000,
    "attribute_5_min": 10,
    "attribute_5_max": 100,

    "copper_price_per_ton": 9500,
    "aluminum_price_per_ton": 2500,
    "copper_price_per_kg": 9.500,    # fixed global Cu price
    "aluminum_price_per_kg": 2.500, # fixed global Al price
}

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regional_data=[
    {
        "region": "United States",
        "copper_product_market_share": 0.65,
        "aluminum_product_market_share": 0.35,
        "weight_attribute_1": 0.30,
        "weight_attribute_2": 0.20,
        "weight_attribute_3": 0.25,
        "weight_attribute_4": 0.12,
        "weight_attribute_5": 0.13
    },
    {

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    "region": "China",
    "copper_product_market_share": 0.30,
    "aluminum_product_market_share": 0.70,
    "weight_attribute_1": 0.40,
    "weight_attribute_2": 0.12,
    "weight_attribute_3": 0.20,
    "weight_attribute_4": 0.10,
    "weight_attribute_5": 0.18
  },
  {
    "region": "India",
    "copper_product_market_share": 0.55,
    "aluminum_product_market_share": 0.45,
    "weight_attribute_1": 0.40,
    "weight_attribute_2": 0.15,
    "weight_attribute_3": 0.20,
    "weight_attribute_4": 0.05,
    "weight_attribute_5": 0.20
  },
  {
    "region": "Europe",
    "copper_product_market_share": 0.40,
    "aluminum_product_market_share": 0.60,
    "weight_attribute_1": 0.22,
    "weight_attribute_2": 0.13,
    "weight_attribute_3": 0.35,
    "weight_attribute_4": 0.15,
    "weight_attribute_5": 0.15
  }
]

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product_data=[
  {
    "region": "United States",
    "dominant_material": "cu",
    "product": "Large Commercial HVAC Fin-and-Tube Heat Exchanger Coil",
    "copper_kg": 600,
    "aluminum_kg": 50,
    "total_mass_kg": 750,
    "non_material_cost_per_unit": 15000,
    "attribute_2_value": 1,
    "attribute_3_value": 78,
    "attribute_4_value": 750,

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"attribute_5_value": 72
},
{
  "region": "United States",
  "dominant material": "al",
  "product": "Large Commercial HVAC Fin-and-Tube Heat Exchanger Coil",
  "copper_kg": 30,
  "aluminum_kg": 380,
  "total_mass_kg": 520,
  "non_material_cost_per_unit": 12000,
  "attribute_2_value": 0,
  "attribute_3_value": 63,
  "attribute_4_value": 520,
  "attribute_5_value": 60
},
{
  "region": "China",
  "dominant material": "cu",
  "product": "Large Commercial HVAC Fin-and-Tube Heat Exchanger Coil",
  "copper_kg": 600,
  "aluminum_kg": 50,
  "total_mass_kg": 750,
  "non_material_cost_per_unit": 8000,
  "attribute_2_value": 1,
  "attribute_3_value": 78,
  "attribute_4_value": 750,
  "attribute_5_value": 72
},
{
  "region": "China",
  "dominant material": "al",
  "product": "Large Commercial HVAC Fin-and-Tube Heat Exchanger Coil",
  "copper_kg": 30,
  "aluminum_kg": 380,
  "total_mass_kg": 520,
  "non_material_cost_per_unit": 6500,
  "attribute_2_value": 0,
  "attribute_3_value": 63,
  "attribute_4_value": 520,
  "attribute_5_value": 60
},
{
  "region": "India",
  "dominant material": "cu",
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"product": "Large Commercial HVAC Fin-and-Tube Heat Exchanger Coil",
"copper_kg": 600,
"aluminum_kg": 50,
"total_mass_kg": 750,
"non_material_cost_per_unit": 7000,
"attribute_2_value": 1,
"attribute_3_value": 78,
"attribute_4_value": 750,
"attribute_5_value": 72
},
{
  "region": "India",
  "dominant material": "al",
  "product": "Large Commercial HVAC Fin-and-Tube Heat Exchanger Coil",
  "copper_kg": 30,
  "aluminum_kg": 380,
  "total_mass_kg": 520,
  "non_material_cost_per_unit": 5500,
  "attribute_2_value": 0,
  "attribute_3_value": 63,
  "attribute_4_value": 520,
  "attribute_5_value": 60
},
{
  "region": "Europe",
  "dominant material": "cu",
  "product": "Large Commercial HVAC Fin-and-Tube Heat Exchanger Coil",
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  "aluminum_kg": 50,
  "total_mass_kg": 750,
  "non_material_cost_per_unit": 16000,
  "attribute_2_value": 1,
  "attribute_3_value": 78,
  "attribute_4_value": 750,
  "attribute_5_value": 72
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "Large Commercial HVAC Fin-and-Tube Heat Exchanger Coil",
  "copper_kg": 30,
  "aluminum_kg": 380,
  "total_mass_kg": 520,
  "non_material_cost_per_unit": 13000,
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"attribute_2_value": 0,  
"attribute_3_value": 63,  
"attribute_4_value": 520,  
"attribute_5_value": 60  
}  
]
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